

Problem Statement and Goals

Squigglr

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Table 1: Revision History

Date	Developer(s)	Change
26 September 2026	Rayan Nasrallah	Added the Problem Statement section, covering the problem, inputs and outputs, stakeholders, and environment, with supporting references.

1 Problem Statement

1.1 Problem

Visual-motor integration is the ability to coordinate visual information with movement. Children rely on it when drawing, copying shapes, and learning to write. Research links these skills to handwriting legibility and identifies motor or handwriting difficulties in some children with attention-deficit/hyperactivity disorder (ADHD) or autism [1–3]. The need extends from kindergarten and primary school learners to older children who require continued support.

Children need opportunities to practise at an appropriate level, understand their difficulties, and develop greater independence. Parents and educators need to distinguish what a child can do independently from what they accomplish with assistance. A finished drawing alone does not provide that distinction. Research on structured foundational activities supports the value of practice for visual-motor development [4].

There is an opportunity to build a system that supports drawing practice suited to individual abilities, provides understandable feedback, and helps chil-

dren and supporting adults track independent progress. The intended outcome is improved drawing skills and greater independence, with opportunities for creative expression. Supporting adults need access to both individual attempt records and understandable summaries of change over time. Clinical diagnosis and treatment are outside the project scope.

1.2 Inputs and Outputs

1.2.1 Inputs

- Selected drawing activities and support preferences.
- Children’s drawing attempts during lessons and creative activities.
- Editing actions, such as moving, resizing, recolouring, or erasing drawing elements.

1.2.2 Outputs

- Drawing targets, demonstrations, and guidance appropriate to the activity and support level.
- Simple activity feedback and completion results based on predefined observations.
- Editable creative artwork, including supported automatic refinements and recoverable original drawings.
- Attempt records showing encountered difficulties, changes within an attempt, guidance used, and the final outcome.
- Progress summaries showing changes across attempts and recurring difficulties, with access to the supporting records.

1.3 Stakeholders

- Kindergarten and primary school children.
- Children of different ages who need additional support, including children with autism, ADHD, or other developmental or learning differences.
- Parents, guardians, and caregivers.
- Educators and learning-support staff.
- School administrators.
- Domain advisers in inclusive education and accessible educational software.
- Dr. Christopher Anand, project supervisor.

1.4 Environment

Users will need access to a supported web browser on a compatible device with a display and a supported drawing input. Using the application will not require installing a separate application.

2 Goals

3 Stretch Goals

[Goals you hope to get to, but not necessary. They are also helpful if the scope of the original project needs to be expanded. —SS]

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

References

1. Daly, C. J., Kelley, G. T., and Krauss, A. (2003). Relationship between visual-motor integration and handwriting skills of children in kindergarten: A modified replication study. *American Journal of Occupational Therapy*, 57(4), 459–462. <https://doi.org/10.5014/ajot.57.4.459>
2. Kaiser, M.-L., Schoemaker, M. M., Albaret, J.-M., and Geuze, R. H. (2015). What is the evidence of impaired motor skills and motor control among children with attention deficit hyperactivity disorder (ADHD)? Systematic review of the literature. *Research in Developmental Disabilities*, 36, 338–357. <https://doi.org/10.1016/j.ridd.2014.09.023>
3. Fuentes, C. T., Mostofsky, S. H., and Bastian, A. J. (2009). Children with autism show specific handwriting impairments. *Neurology*, 73(19), 1532–1537. <https://doi.org/10.1212/WNL.0b013e3181c0d48c>
4. Taverna, L., Tremolada, M., Dozza, L., Zanin Scaratti, R., Domahs, U., Lallo, C., and Toso, B. (2020). Who benefits from an intervention program on foundational skills for handwriting addressed to kindergarten children and first graders? *International Journal of Environmental Research and Public Health*, 17(6), 2166. <https://doi.org/10.3390/ijerph17062166>